

Tutorial 2

1. Consider the slotted transmission protocol. There are N nodes in the network. Each node transmits the packet with probability p in each slot (same probability for all nodes). Derive the optimal value of p such that the sum of throughput of all nodes in the network per slot of this slotted transmission network is maximized.
2. The RTS-CTS mechanism is designed to address the problem of hidden terminal and exposed terminal. Can you think of a scenario when the hidden terminal problem will remain in spite of RTS-CTS?
3. Explain how energy consumption and delay can be traded off in a wireless network. Can you use this trade-off in a distributed environment? Why or why not?
4. Discuss how the contention window size for backoff affects the network performance.